

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

**Product Name** Mercury standard solution, 1 mg/ml Hg in 2 to 5% HNO<sub>3</sub>

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>196120000; 196121000; 196125000</b>                                                      |
| <b>Address</b>                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>09 980 6780 or +64 9 980 6780</b>                                    |
| <b>Telephone / Fax Numbers</b> | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| <b>E-mail address</b>          | <a href="mailto:ANZinfo@thermofisher.com">ANZinfo@thermofisher.com</a>                      |

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

### GHS Classification

#### Physical hazards

Substances/mixtures corrosive to metal Category 1

#### Health hazards

Skin Corrosion/Irritation Category 2  
 Serious Eye Damage/Eye Irritation Category 1  
 Specific target organ toxicity - (repeated exposure) Category 2

#### Environmental hazards

Based on available data, the classification criteria are not met

### Label Elements



Signal Word

Danger

#### Hazard Statements

H290 - May be corrosive to metals

H315 - Causes skin irritation

H318 - Causes serious eye damage

H373 - May cause damage to organs through prolonged or repeated exposure

#### Precautionary Statements

##### Prevention

P234 - Keep only in original packaging

P264 - Wash face, hands and any exposed skin thoroughly after handling

P280 - Wear protective gloves/protective clothing/eye protection/face protection

##### Response

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water

P310 - Immediately call a POISON CENTER or doctor

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P390 - Absorb spillage to prevent material damage

P362 + P364 - Take off contaminated clothing and wash it before reuse

##### Storage

P406 - Store in corrosion resistant polypropylene container with a resistant inliner

##### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

Other hazards which do not result in classification

## Section 3 - Composition and Information on Ingredients

| Component                   | CAS No    | Weight % |
|-----------------------------|-----------|----------|
| Nitric acid ...% [C ≤ 70 %] | 7697-37-2 | 2-5      |
| Water                       | 7732-18-5 | 95-98    |

## Section 4 - First Aid Measures

### Description of first aid measures

#### General Advice

If symptoms persist, call a physician.

#### New Zealand Emergency Tel.

CHEMTREC®  
09 980 6780 or +64 9 980 6780

#### Inhalation

If not breathing, give artificial respiration. Remove to fresh air. Get medical attention if symptoms occur.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists,

---

|                                            |                                                                                                                                                                                                                                                                                         |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | call a physician.                                                                                                                                                                                                                                                                       |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water.                                                                                                                                                                                                                            |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                        |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                    |
| <b>Most important symptoms and effects</b> | None reasonably foreseeable. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                                                                                                                                                                  |

## **Section 5 - Fire Fighting Measures**

### **Suitable Extinguishing Media**

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

### **Extinguishing media which must not be used for safety reasons**

No information available.

### **Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes.

### **Hazardous Combustion Products**

Nitrogen oxides (NO<sub>x</sub>), Thermal decomposition can lead to release of irritating gases and vapors.

### **Special protective equipment and precautions for fire fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## **Section 6 - Accidental Release Measures**

### **Personal Precautions, Protective Equipment and Emergency Procedures**

#### **Emergency procedures**

Ensure adequate ventilation. Use personal protective equipment as required.

#### **Environmental Precautions**

Should not be released into the environment.

#### **Methods for Containment and Clean Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

#### **Precautions to prevent secondary hazards**

Clean contaminated objects and areas thoroughly observing environmental regulations

#### **Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## **Section 7 - Handling and Storage**

### **Precautions for Safe Handling**

#### **Advice on safe handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Ensure adequate ventilation. Avoid

ingestion and inhalation.

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

#### Conditions for Safe Storage, Including any Incompatibilities

##### Storage Conditions

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Do not store in metal containers.

##### Incompatible Materials

Strong oxidizing agents. Bases. Metals.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

#### Control parameters

##### Exposure limits

**NZ** - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

**ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

| Component                   | New Zealand WEL                                                                       | Australia                                                                             | ACGIH TLV                 | The United Kingdom                                       |
|-----------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------|----------------------------------------------------------|
| Nitric acid ...% [C ≤ 70 %] | TWA: 2 ppm<br>TWA: 5.2 mg/m <sup>3</sup><br>STEL: 4 ppm<br>STEL: 10 mg/m <sup>3</sup> | STEL: 4 ppm<br>STEL: 10 mg/m <sup>3</sup><br>TWA: 2 ppm<br>TWA: 5.2 mg/m <sup>3</sup> | TWA: 2 ppm<br>STEL: 4 ppm | STEL: 1 ppm 15 min<br>STEL: 2.6 mg/m <sup>3</sup> 15 min |

##### Biological limit values

**ACGIH** - American Conference of Governmental Industrial Hygienists (ACGIH) TLVs® and BEIs®- Threshold Limit Values for Chemical Substances and Physical Agents & Biological Exposure Indices. 2022 Edition

#### Appropriate engineering controls

##### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

#### Individual protection measures, such as personal protective equipment

##### Eye Protection

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

##### Hand Protection

Protective gloves

| Glove material                                               | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|--------------------------------------------------------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Natural rubber, Nitrile rubber, Neoprene, PVC, Butyl rubber. | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g.

sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                        |                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b>        | Long sleeved clothing                                                                                                                                                                                                                                                                                                       |
| <b>Respiratory Protection</b>          | Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices |
| <b>Recommended Filter type:</b>        | Particulates filter conforming to EN 143 Acid gases filter Type E Yellow conforming to EN14387 (or AUS/NZ equivalent)                                                                                                                                                                                                       |
| <b>Recommended half mask:-</b>         | Particle filtering: EN149:2001 Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)<br>When RPE is used a face piece Fit Test should be conducted                                                                                                                                          |
| <b>Hygiene Measures</b>                | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                                                                                                      |
| <b>Environmental exposure controls</b> | No information available.                                                                                                                                                                                                                                                                                                   |

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Physical State</b>                          | Liquid                   |                                          |
| <b>Appearance</b>                              | No information available |                                          |
| <b>Odor</b>                                    | Odorless                 |                                          |
| <b>Odor Threshold</b>                          | No data available        |                                          |
| <b>pH</b>                                      | < 1                      |                                          |
| <b>Melting Point/Range</b>                     | No data available        |                                          |
| <b>Softening Point</b>                         | No data available        |                                          |
| <b>Boiling Point/Range</b>                     | 101 °C / 213.8 °F        |                                          |
| <b>Flammability (liquid)</b>                   | No data available        |                                          |
| <b>Flammability (solid,gas)</b>                | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Flash Point</b>                             | No information available | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | No data available        |                                          |
| <b>Decomposition Temperature</b>               | No data available        |                                          |
| <b>Viscosity</b>                               | No data available        |                                          |
| <b>Water Solubility</b>                        | Soluble                  |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Component</b>                               | <b>log Pow</b>           |                                          |
| Nitric acid ...% [C ≤ 70 %]                    | -2.3                     |                                          |
| <b>Vapor Pressure</b>                          | No data available        |                                          |
| <b>Density / Specific Gravity</b>              | 1.02                     |                                          |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                   |
| <b>Vapor Density</b>                           | No data available        | (Air = 1.0)                              |
| <b>Particle characteristics</b>                | Not applicable (liquid)  |                                          |

### Other information

## Section 10 - Stability and Reactivity

|                   |                                            |
|-------------------|--------------------------------------------|
| <b>Reactivity</b> | None known, based on information available |
| <b>Stability</b>  | Stable under normal conditions.            |

|                                         |                                                                                                               |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                      |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                      |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                      |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                                 |
| <b>Conditions to Avoid</b>              | Incompatible products, Excess heat.                                                                           |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Bases, Metals.                                                                       |
| <b>Hazardous Decomposition Products</b> | Nitrogen oxides (NO <sub>x</sub> ). Thermal decomposition can lead to release of irritating gases and vapors. |

## Section 11 - Toxicological Information

### Acute Effects

### Information on likely routes of exposure

#### Product Information

|                   |                                                                                                 |
|-------------------|-------------------------------------------------------------------------------------------------|
| <b>Inhalation</b> | Not an expected route of exposure.                                                              |
| <b>Eyes</b>       | Avoid contact with eyes. Corrosive to the eyes and may cause severe damage including blindness. |
| <b>Skin</b>       | Avoid contact with skin. Causes burns. Skin Corrosion/Irritation.                               |
| <b>Ingestion</b>  | May be harmful if swallowed.                                                                    |

### Numerical measures of toxicity

#### (a) acute toxicity;

|                   |                   |
|-------------------|-------------------|
| <b>Oral</b>       | No data available |
| <b>Dermal</b>     | No data available |
| <b>Inhalation</b> | No data available |

### Toxicology data for the components

| Component                   | LD50 Oral | LD50 Dermal | LC50 Inhalation           |
|-----------------------------|-----------|-------------|---------------------------|
| Nitric acid ...% [C ≤ 70 %] |           |             | LC50 = 2500 ppm. (Rat) 1h |
| Water                       | -         | -           | -                         |

| Component                   | ECHA (RAC) ATE (Oral) | ECHA (RAC) ATE (Dermal) | ECHA (RAC) ATE (Inhalation) |
|-----------------------------|-----------------------|-------------------------|-----------------------------|
| Nitric acid ...% [C ≤ 70 %] | -                     | -                       | ATE = 2.65 mg/L (vapours)   |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

#### (d) respiratory or skin sensitization;

|                    |                   |
|--------------------|-------------------|
| <b>Respiratory</b> | No data available |
| <b>Skin</b>        | No data available |

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs Urinary Tract.

(j) aspiration hazard; No data available

**Symptoms / effects, both acute and delayed**

Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

## Section 12 - Ecological Information

### Ecotoxicity

Aquatic ecotoxicity Do not empty into drains. .

Terrestrial ecotoxicity There is no data for this product

### **Persistence and Degradability**

Persistence Miscible with water, Persistence is unlikely, based on information available, Soluble in water.

Bioaccumulative Potential Bioaccumulation is unlikely

| Component                   | log Pow | Bioconcentration factor (BCF) |
|-----------------------------|---------|-------------------------------|
| Nitric acid ...% [C ≤ 70 %] | -2.3    | No data available             |

Mobility The product is water soluble, and may spread in water systems. . Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

### Other adverse effects

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors

Persistent Organic Pollutant This product does not contain any known or suspected substance

Ozone Depletion Potential This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

### Waste treatment methods

Waste from Residues/Unused Products Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

|                               |                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Contaminated Packaging</b> | Dispose of this container to hazardous or special waste collection point.                                                                                                                                                                                                                                                                                                                             |
| <b>Other Information</b>      | Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations . Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Solutions with low pH-value must be neutralized before discharge. Do not flush to sewer. Large amounts will affect pH and harm aquatic organisms. |

## **Section 14 - Transport Information**

| <b>Component</b>                                 | <b>Hazchem Code</b> |
|--------------------------------------------------|---------------------|
| Nitric acid ...% [C ≤ 70 %]<br>7697-37-2 ( 2-5 ) | 2R<br>2P<br>2PE     |

### **NZS 5433:2020**

|                                |                                             |
|--------------------------------|---------------------------------------------|
| <b>UN-No</b>                   | UN3264                                      |
| <b>Proper Shipping Name</b>    | Corrosive liquid, acidic, inorganic, n.o.s. |
| <b>Technical Shipping Name</b> | Nitric acid                                 |
| <b>Hazard Class</b>            | 8                                           |
| <b>Packing Group</b>           | III                                         |

### **IATA**

|                                |                                             |
|--------------------------------|---------------------------------------------|
| <b>UN-No</b>                   | UN3264                                      |
| <b>Proper Shipping Name</b>    | Corrosive liquid, acidic, inorganic, n.o.s. |
| <b>Technical Shipping Name</b> | Nitric acid                                 |
| <b>Hazard Class</b>            | 8                                           |
| <b>Packing Group</b>           | III                                         |

### **IMDG/IMO**

|                                |                                             |
|--------------------------------|---------------------------------------------|
| <b>UN-No</b>                   | UN3264                                      |
| <b>Proper Shipping Name</b>    | Corrosive liquid, acidic, inorganic, n.o.s. |
| <b>Technical Shipping Name</b> | Nitric acid                                 |
| <b>Hazard Class</b>            | 8                                           |
| <b>Packing Group</b>           | III                                         |

|                              |                       |
|------------------------------|-----------------------|
| <b>Environmental hazards</b> | No hazards identified |
|------------------------------|-----------------------|

|                                                                                 |                                |
|---------------------------------------------------------------------------------|--------------------------------|
| <b>Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code</b> | Not applicable, packaged goods |
|---------------------------------------------------------------------------------|--------------------------------|

|                            |                                                                                                                         |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Special Precautions</b> | No special precautions required. Please refer to the applicable dangerous goods regulations for additional information. |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------|

|                               |            |
|-------------------------------|------------|
| <b>Additional information</b> | None known |
|-------------------------------|------------|

## **Section 15 - Regulatory Information**

### **Safety, health and environmental regulations/legislation specific for the substance or mixture**

#### **National Regulations**

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

#### **Certified handlers, tracking and controlled substance license requirements**

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most

explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

#### Prohibition or notification/licensing requirements

Shown below are details of specific prohibition/notifications or licensing requirements when they apply.

#### International Regulations

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Chemicals Subject to Prior Informed Consent (PIC)

#### Authorisation/Restrictions according to EU REACH

| Component                   | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Nitric acid ...% [C ≤ 70 %] | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

#### International Inventories

New Zealand (NZIoC), Australia (AICS), Europe (EINECS/ELINCS/NLP), Korea (KECL), China (IECSC), Taiwan (TCSI), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component                   | CAS No    | NZIoC | AICS | EINECS    | ELINCS | NLP | KECL     | IECSC | TCSI |
|-----------------------------|-----------|-------|------|-----------|--------|-----|----------|-------|------|
| Nitric acid ...% [C ≤ 70 %] | 7697-37-2 | X     | X    | 231-714-2 | -      | -   | KE-25911 | X     | X    |
| Water                       | 7732-18-5 | X     | X    | 231-791-2 | -      | -   | KE-35400 | X     | X    |

| Component                   | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | PICCS | ISHL | ENCS |
|-----------------------------|-----------|------|-----------------------------------------------|-----|------|-------|------|------|
| Nitric acid ...% [C ≤ 70 %] | 7697-37-2 | X    | ACTIVE                                        | X   | -    | X     | X    | X    |
| Water                       | 7732-18-5 | X    | ACTIVE                                        | X   | -    | X     | -    | X    |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Section 16 - Other Information

**This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations**

#### Legend

**NZIoC** - New Zealand Inventory of Chemicals

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**IECSC** - Chinese Inventory of Existing Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**TWA** - Time Weighted Average

**AICS** - Australian Inventory of Chemical Substances

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**CAS** - Chemical Abstracts Service

**ACGIH** - American Conference of Governmental Industrial Hygienists

**IARC** - International Agency for Research on Cancer  
**NZS 5433:2020** - Transport of Dangerous Goods on Land  
**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association  
**MARPOL** - International Convention for the Prevention of Pollution from Ships  
**LD50** - Lethal Dose 50%  
**EC50** - Effective Concentration 50%  
**WEL** - Workplace Exposure Limit  
**DNEL** - Derived No Effect Level  
**POW** - Partition coefficient Octanol:Water  
**vPvB** - very Persistent, very Bioaccumulative  
**VOC** - (Volatile Organic Compound)

**PNEC** - Predicted No Effect Concentration  
**OECD** - Organisation for Economic Co-operation and Development  
**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code  
**LC50** - Lethal Concentration 50%  
**ATE** - Acute Toxicity Estimate  
**RPE** - Respiratory Protective Equipment  
**NOEC** - No Observed Effect Concentration  
**BCF** - Bioconcentration factor  
**PBT** - Persistent, Bioaccumulative, Toxic

#### Key literature references and sources for data

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).  
<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS  
EPA Guide to classifying hazardous substances in New Zealand  
EPA - Assigning a product to an existing HSNO approval guide

#### Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

|                              |                       |
|------------------------------|-----------------------|
| <b>Physical hazards</b>      | On basis of test data |
| <b>Health Hazards</b>        | Calculation method    |
| <b>Environmental hazards</b> | Calculation method    |

#### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.  
Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.  
First aid for chemical exposure, including the use of eye wash and safety showers.  
Chemical incident response training.

|                         |                |
|-------------------------|----------------|
| <b>Revision Date</b>    | 10-Mar-2023    |
| <b>Revision Summary</b> | Not applicable |

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of Safety Data Sheet